Macroeconomics I — MPE Insper

Practice list: Measurement, Welfare and Growth — graduate level (Listas 1–2)

Solutions and grading rubric

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Block 1 — Price indices, PPP and welfare across countries

(Kurlat chs. 1–2 • 25 points)

Question 1. (10 points) Objective items. Each item needs a short derivation; write it on scratch paper.

a) (3 points) A household has preferences $u(x_1, x_2) = x_1^{1/3} x_2^{2/3}$ over electricity (x_1) and everything else (x_2). Between two months the price of electricity rises 44% and the other price does not change. The true cost-of-living (Konüs) index, defined as the ratio of minimum expenditures needed to reach the *same* utility at the new and old prices, records inflation of:

A. 14.67%

B. 12.92%

C. 12.99%

D. 11.34%

Answer: B

Step — expenditure function. $\min p_1 x_1 + p_2 x_2$ s.t. $x_1^{1/3} x_2^{2/3} \geq \bar{u}$. The FOCs give $x_1 = \frac{1}{3} e/p_1$, $x_2 = \frac{2}{3} e/p_2$ (constant expenditure shares $s_1 = \frac{1}{3}$, $s_2 = \frac{2}{3}$). Substituting into the constraint: $e(p, \bar{u}) = \bar{u}\,(3p_1)^{1/3}(\frac{3}{2}p_2)^{2/3}$.

Step — Konüs index. $P^* = \dfrac{e(p^1, \bar{u})}{e(p^0, \bar{u})} = \left(\dfrac{p_1^1}{p_1^0}\right)^{1/3}\left(\dfrac{p_2^1}{p_2^0}\right)^{2/3} = 1.44^{1/3} = 1.1292$: the geometric mean of the price relatives, weighted by the shares, independent of $\bar{u}$ (homotheticity).

Step — distractors. With constant shares, Laspeyres $= \sum_i s_i R_i = \frac{1}{3}1.44 + \frac{2}{3} = 1.1467$ (A); Paasche $= (\sum_i s_i/R_i)^{-1} = 1.1134$ (D); Fisher $= \sqrt{1.1467 \cdot 1.1134} = 1.1299$ (C).

Ref: Kurlat (2020), §1.2, pp. 22–25; Jones (2020), §2.3, pp. 32–33.

b) (3 points) Brazil and the US produce a tradable good with $Y_T = A_T L_T$ and a non-tradable service with $Y_N = A_N L_N$. Labor is mobile across sectors within each country, markets are competitive, the tradable obeys the law of one price, and consumers' price index is $P = p_T^{1/2} p_N^{1/2}$. Brazil's productivity relative to the US is 0.25 in tradables and 0.80 in non-tradables. Brazil's price level relative to the US, both measured in dollars at the market exchange rate, is:

A. 0.250

B. 0.800

C. 0.559

D. 0.313

Answer: C

Step — zero profits. With a common wage W: $p_T = W/A_T$, $p_N = W/A_N$, so $p_N/p_T = A_T/A_N$.

Step — price level. $P = p_T(p_N/p_T)^{1/2} = p_T(A_T/A_N)^{1/2}$. The law of one price sets $p_T^{BR} = E\,p_T^{US}$, hence $\dfrac{P^{BR}}{E\,P^{US}} = \left[\dfrac{(A_T/A_N)^{BR}}{(A_T/A_N)^{US}}\right]^{1/2} = \left(\dfrac{0.25}{0.80}\right)^{1/2} = 0.559$. Option D forgets the share 1/2.

Economic intuition. Low tradable productivity means low wages; low wages make services cheap. Market-rate GDP therefore understates Brazil's real output by a factor $1/0.559 \approx 1.79$ (for a GDP basket like the consumption basket).

Ref: Kurlat (2020), §1.2, pp. 24–27; Romer (2012), §4.2, eq. (4.16), pp. 160–161.

c) (2 points, T or F) With Cobb–Douglas preferences the Fisher index equals the true cost-of-living index, since it is the geometric mean of an upper bound (Laspeyres) and a lower bound (Paasche).

Answer: F

Step — three means. With constant shares, Laspeyres is the arithmetic mean (AM), Paasche the harmonic mean (HM) and the true index the geometric mean (GM) of the price relatives. HM $\leq$ GM $\leq$ AM is true, but $\sqrt{\text{AM} \cdot \text{HM}} = \text{GM}$ only in special cases (two goods with equal shares, where AM·HM$= R_1 R_2$). In (a): Fisher $12.99\% \neq 12.92\%$.

Ref: Kurlat (2020), §1.2.

d) (2 points, T or F) If log consumption is normal with variance v and utility is CRRA with curvature σ, the consumption-equivalent cost of inequality, $1 - e^{-\sigma v/2}$, depends neither on mean consumption nor on the value of life $\bar{u}$, while the welfare cost of shorter life expectancy depends on both.

Answer: T

Step — why. Question 2(a) proves $c_{CE} = \bar{c}\, e^{-\sigma v/2}$: a proportional factor, free of $\bar{c}$ and $\bar{u}$. Life expectancy multiplies the whole flow utility $\bar{u} + u(c_{CE}) - \theta(1-l)^2$, which depends on both (Question 2(b)).

Ref: Kurlat (2020), §2.2, eqs. (2.2.1)–(2.2.5).

Question 2. (15 points) Discursive. The Jones–Klenow measure (Kurlat §2.2). "Rawls" is placed at random in country $j \in \{BR, US\}$. He is alive with probability a_j (life expectancy over 100) and, if alive, gets $\bar{u} + u(c) - \theta(1-l_j)^2$, where $1 - l_j$ is market work (the same for everyone) and $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$ ($\ln c$ if $\sigma = 1$). Consumption is lognormal: $\ln c \sim N(m_j, v_j)$, with mean $\bar{c}_j = e^{m_j + v_j/2}$. The welfare of Brazil relative to the US is the λ that makes Rawls indifferent between Brazil and a US in which *every* consumption level is multiplied by λ. Data: $\bar{c}_{US} = 1$, $\bar{c}_{BR} = 0.30$, $v_{US} = 0.4$, $v_{BR} = 1.0$, $a_{US} = 0.79$, $a_{BR} = 0.76$, $1 - l_{US} = 0.20$, $1 - l_{BR} = 0.22$, $\theta = 2$, $\bar{u} = 5$.

a) (3 points) Prove that $\mathbb{E}[u(c)] = u(c_{CE})$ with $c_{CE} = \bar{c}\, e^{-\sigma v/2}$ for every $\sigma > 0$, including $\sigma = 1$. Interpret the factor $e^{-\sigma v/2}$.

Step — normal moment generating function. If $X \sim N(m, v)$ then $\mathbb{E}[e^{tX}] = e^{tm + t^2 v/2}$ for every real t.

Step — case $\sigma \neq 1$. With $X = \ln c$ and $t = 1 - \sigma$:

$$\mathbb{E}[c^{1-\sigma}] = e^{(1-\sigma)m + (1-\sigma)^2 v/2}.$$

Substitute $m = \ln \bar{c} - v/2$:

$$\mathbb{E}[c^{1-\sigma}] = \bar{c}^{\,1-\sigma} e^{-(1-\sigma)v/2 + (1-\sigma)^2 v/2} = \bar{c}^{\,1-\sigma} e^{-\sigma(1-\sigma)v/2} = \left(\bar{c}\, e^{-\sigma v/2}\right)^{1-\sigma}.$$

Since u is affine in $c^{1-\sigma}$, $\mathbb{E}[u(c)] = u(\bar{c}\, e^{-\sigma v/2})$.

Step — case $\sigma = 1$. $\mathbb{E}[\ln c] = m = \ln \bar{c} - v/2 = \ln(\bar{c}\, e^{-v/2})$: the same formula.

Step — interpretation. Lognormal inequality acts as a proportional tax $e^{-\sigma v/2}$ on mean consumption. It grows with dispersion v and with curvature σ: behind the veil of ignorance, inequality is risk, and σ is risk aversion. It does not depend on $\bar{c}$.

Ref: Kurlat (2020), §2.2, eqs. (2.2.4)–(2.2.6), pp. 35–37.

Grading rubric.

- Uses the moment generating function correctly: 1 pt.
- Algebra to $(\bar{c}e^{-\sigma v/2})^{1-\sigma}$ and the affine step: 1 pt.
- Log case and interpretation: 1 pt. Stating the result without proof: max. 1 pt in the item.

b) (4 points) Let $\sigma = 1$. Derive λ and an *exact* additive decomposition of $\ln \lambda$ into consumption, inequality, leisure and life-expectancy terms. Compute every term.

Step — expected utility. Scaling every consumption level by λ adds $\ln \lambda$ to $\ln c$. Using (a), with $h_j \equiv 1 - l_j$:

$$U_j(\lambda) = a_j\big[\bar{u} + \ln \lambda + \ln \bar{c}_j - \tfrac{v_j}{2} - \theta h_j^2\big].$$

Step — indifference. $U_{US}(\lambda) = U_{BR}(1)$. Define $X_{BR} \equiv \bar{u} + \ln \bar{c}_{BR} - v_{BR}/2 - \theta h_{BR}^2$ (flow utility of an average-risk Brazilian). Then

$$\ln \lambda = \frac{a_{BR}}{a_{US}} X_{BR} - \bar{u} - \ln \bar{c}_{US} + \frac{v_{US}}{2} + \theta h_{US}^2.$$

Step — exact decomposition. Add and subtract X_{BR}:

$$\ln \lambda = \underbrace{\ln \frac{\bar{c}_{BR}}{\bar{c}_{US}}}_{\text{consumption}} \underbrace{- \frac{v_{BR} - v_{US}}{2}}_{\text{inequality}} \underbrace{- \theta\big(h_{BR}^2 - h_{US}^2\big)}_{\text{leisure}} + \underbrace{\Big(\frac{a_{BR}}{a_{US}} - 1\Big) X_{BR}}_{\text{life expectancy}}.$$

No approximation is involved.

Step — numbers. Consumption $\ln 0.30 = -1.20397$; inequality -0.30000; leisure $-2(0.0484 - 0.0400) = -0.01680$; $X_{BR} = 5 - 1.20397 - 0.5 - 0.0968 = 3.19923$; life expectancy $(0.76/0.79 - 1) \cdot 3.19923 = -0.12149$. Total $\ln \lambda = -1.64226$, so $\lambda = 0.1936$.

Economic intuition. Brazil's welfare is about 19% of the US level, against a consumption ratio of 30%: inequality removes 0.30 log points, shorter lives 0.12, longer hours 0.02.

Ref: Kurlat (2020), §2.2, eqs. (2.2.1)–(2.2.2), pp. 33–40.

Grading rubric.

- Expected utility with $\mathbb{E} \ln c = \ln \bar{c} - v/2$ and a_j multiplying the whole bracket: 1 pt.
- Solves for $\ln \lambda$: 1 pt.
- Exact decomposition: 1 pt. A decomposition that leaves an unexplained residual: max. 0.5 pt on this line.
- Numbers: 1 pt (-1 pt for an arithmetic slip with the right method).

c) (4 points) Now let σ be general. Derive λ in closed form. Prove that it tends to the formula of (b) as $\sigma \to 1$. Prove that the pure inequality factor (all else equal across countries) falls with σ when $v_{BR} > v_{US}$, and give its limits as $\sigma \to 0$ and $\sigma \to \infty$. Explain why Kurlat's unnormalized $c^{1-\sigma}/(1-\sigma)$ with a fixed $\bar{u}$ has no limit at $\sigma = 1$.

Step — indifference. By (a), scaling every consumption by λ scales c_{CE} by λ: $U_{US}(\lambda) = a_{US}[\bar{u} + u(\lambda c_{CE}^{US}) - \theta h_{US}^2]$. Indifference with Brazil:

$$u(\lambda c_{CE}^{US}) = Z \equiv \frac{a_{BR}}{a_{US}}\big[\bar{u} + u(c_{CE}^{BR}) - \theta h_{BR}^2\big] - \bar{u} + \theta h_{US}^2.$$

Step — inversion. $u^{-1}(Z) = [1 + (1-\sigma)Z]^{1/(1-\sigma)}$, so

$$\lambda = \frac{[1 + (1-\sigma)Z]^{1/(1-\sigma)}}{c_{CE}^{US}}, \qquad c_{CE}^{j} = \bar{c}_j e^{-\sigma v_j/2},$$

defined when $1 + (1-\sigma)Z > 0$.

Step — limit $\sigma \to 1$. By L'Hôpital in σ, $\frac{c^{1-\sigma}-1}{1-\sigma} \to \ln c$, so $Z \to Z_{\log}$; and $[1+(1-\sigma)Z]^{1/(1-\sigma)} \to e^Z$ (the definition of e). Hence $\lambda \to e^{Z_{\log}}/c_{CE}^{US}$, which is exactly the formula of (b).

Step — inequality factor. With equal a, h and $\bar{c}$, $\lambda = c_{CE}^{BR}/c_{CE}^{US} = e^{-\sigma(v_{BR}-v_{US})/2}$, and $\partial\lambda/\partial\sigma = -\frac{v_{BR}-v_{US}}{2}\lambda < 0$. As $\sigma \to 0$ the factor $\to 1$ (risk neutrality: inequality is irrelevant); as $\sigma \to \infty$ it $\to 0$ (a maximin, Rawlsian judgment; lognormal consumption has no positive lower bound).

Step — normalization. $\frac{c^{1-\sigma}}{1-\sigma} = \frac{c^{1-\sigma}-1}{1-\sigma} + \frac{1}{1-\sigma}$. The constant $1/(1-\sigma)$ diverges at $\sigma = 1$; with $\bar{u}$ held fixed it changes the value of being alive by an unbounded amount, so λ explodes. The value of life must be recalibrated whenever σ changes; normalizing u is one consistent way.

Ref: Kurlat (2020), §2.2, footnote 6 and Exercise 2.10 (Rawlsian limit).

Grading rubric.

- Indifference condition and inversion: 1.5 pts.
- Limit at $\sigma = 1$ with an argument (L'Hôpital or series): 1 pt. Asserting it: 0 for this line.
- Sign of $\partial\lambda/\partial\sigma$ and both limits: 1 pt.
- Normalization argument: 0.5 pt.

d) (4 points) Compute the inequality factor at $\sigma = 1$ and $\sigma = 2$. Then show that with $\sigma = 2$ and $\bar{u} = 5$ the flow utility of the Brazilian certainty-equivalent resident is negative, and explain why that makes λ meaningless. What must the analyst do?

Step — factors. $e^{-1\cdot 0.6/2} = 0.7408$ and $e^{-2\cdot 0.6/2} = 0.5488$: the cost of Brazil's extra inequality rises from 26% to 45% of consumption.

Step — negative flow utility. $c_{CE}^{BR} = 0.30\, e^{-1.0} = 0.1104$; $u = 1 - 1/0.1104 = -8.061$; flow utility $5 - 8.061 - 2(0.22)^2 = -3.158 < 0$.

Step — why it breaks. With negative flow utility, being alive is worse than being dead (who gets zero). A higher survival probability a_{BR} would *lower* welfare, and λ would reward shorter lives.

Step — remedy. Recalibrate $\bar{u}$ jointly with σ (Jones–Klenow match the value of a statistical life), so that flow utility is positive over the relevant consumption range.

Economic intuition. σ does double duty: it is intertemporal curvature, risk aversion and aversion to inequality at once. Welfare rankings that move with σ should be reported with it.

Ref: Kurlat (2020), §2.2, "The value of life expectancy", pp. 38–39; Exercise 2.6.

Grading rubric.

- Both factors: 1 pt.
- Negative flow utility computed: 1 pt.
- The perverse effect of life expectancy explained: 1 pt.
- Recalibration of $\bar{u}$: 1 pt.

Block 2 — Solow: CES, exact dynamics and human capital

(Kurlat chs. 4–5; Romer ch. 1 • 25 points)

Question 3. (10 points) Objective items. Work *per effective worker*: $\tilde{k} = K/(AL)$, $\dot{\tilde{k}} = sf(\tilde{k}) - (n+g+\delta)\tilde{k}$ in continuous time.

a) (3 points) Let $f(\tilde{k}) = [\alpha\tilde{k}^{\rho} + 1 - \alpha]^{1/\rho}$ with $\alpha = 1/3$, $\rho = 1/2$ (elasticity of substitution 2), and $n + g + \delta = 0.08$. The saving rate above which $\tilde{k}$ grows forever, with no steady state, is:

A. 24%

B. 33.3%

C. 72%

D. none: the Solow model always has a steady state

Answer: C

Step — growth rate. $\dot{\tilde{k}}/\tilde{k} = s\, f(\tilde{k})/\tilde{k} - (n+g+\delta)$, and $f(k)/k = [\alpha + (1-\alpha)k^{-\rho}]^{1/\rho}$ falls from ∞ ($k \to 0$) to $\alpha^{1/\rho} = 1/9$ ($k \to \infty$).

Step — threshold. A steady state exists iff $s\alpha^{1/\rho} < n + g + \delta$, i.e. $s < 0.08 \cdot 9 = 72\%$. Above it, $\tilde{k}$ grows asymptotically at $s/9 - 0.08 > 0$: an AK limit. Option A uses α instead of $\alpha^{1/\rho}$; D assumes the Inada conditions, which fail when $\rho > 0$.

Ref: Romer (2012), §1.8, pp. 42–43; Kurlat (2020), Exercise 4.4.

b) (3 points) Let $f(\tilde{k}) = \tilde{k}^{1/3}$ and $n + g + \delta = 0.08$. The economy starts at $\tilde{k}(0) = \tilde{k}^*/8$. The time until $\tilde{k}$ reaches $\tilde{k}^*/2$ is:

A. 13.0 years

B. 20.6 years

C. 13.2 years

D. 6.6 years

Answer: C

Step — exact solution. Let $z \equiv \tilde{k}^{1-\alpha}$ $(= K/Y)$. Then $\dot{z} = (1-\alpha)\tilde{k}^{-\alpha}\dot{\tilde{k}} = (1-\alpha)[s-(n+g+\delta)z]$, a linear ODE: $z(t) = z^* + (z(0) - z^*)e^{-\lambda t}$, $\lambda = (1-\alpha)(n+g+\delta) = 0.0533$.

Step — timing. $z/z^* = (\tilde{k}/\tilde{k}^*)^{2/3}$: $z(0)/z^* = 1/4$. Solve $1 - 0.75e^{-\lambda t} = 0.5^{2/3} = 0.6300$: $e^{-\lambda t} = 0.4934$, $t = 0.7065/0.0533 = 13.2$ years.

Step — distractors. B is the log-linear approximation ($\ln(\tilde{k}/\tilde{k}^*)$ shrinking at rate λ: $e^{-\lambda t} = \ln 2/\ln 8$, $t = 20.6$), poor this far from the steady state; A is the half-life $\ln 2/\lambda$.

Ref: Romer (2012), §1.5, eqs. (1.27)–(1.31), pp. 24–26.

c) (2 points, T or F) With CES production and an elasticity of substitution below one ($\rho < 0$), the capital share $\alpha_K(\tilde{k}) = f'(\tilde{k})\tilde{k}/f(\tilde{k})$ falls as $\tilde{k}$ rises.

Answer: T

Step — share. $\alpha_K = \dfrac{\alpha k^\rho}{\alpha k^\rho + 1 - \alpha} = \left[1 + \frac{1-\alpha}{\alpha}k^{-\rho}\right]^{-1}$; with $\rho < 0$, $k^{-\rho}$ rises with k, so α_K falls: with complements, the abundant factor's share shrinks. The convergence speed, $(1 - \alpha_K(\tilde{k}^*))(n+g+\delta)$, rises with $\tilde{k}^*$.

Ref: Romer (2012), §1.5 and §1.8.

d) (2 points, T or F) On a balanced growth path with $Y = K^\alpha(AL)^{1-\alpha}$ and A growing at g, growth accounting attributes all of the growth of output per worker to the Solow residual.

Answer: F

Step — accounting. On the path $g_K = g_Y = n + g$, $g_L = n$: residual $= g_Y - \alpha g_K - (1-\alpha)g_L = (1-\alpha)g$; capital deepening $\alpha(g_K - g_L) = \alpha g$. Accounting assigns a share α to capital, although technology causes all of it: capital deepening is induced by A.

Ref: Kurlat (2020), §5.4, eq. (5.4.2), p. 87; Romer (2012), problem 1.13.

Question 4. (15 points) Discursive. Mankiw–Romer–Weil with Brazilian data. Output is

$$Y = K^\alpha H^\beta (AL)^{1-\alpha-\beta}, \qquad \alpha + \beta < 1,$$

with $\dot{K} = s_k Y - \delta K$ and $\dot{H} = s_h Y - \delta H$; L and A grow at n and g. Let $k = K/(AL)$, $h = H/(AL)$, $y = k^\alpha h^\beta$ and $N \equiv n + g + \delta$.

a) (3 points) Derive the laws of motion of k and h, the steady state (k^*, h^*) and $\ln y^*$. Give the elasticity of y^* with respect to s_k.

Step — laws of motion. $\dot{k}/k = \dot{K}/K - n - g = s_k y/k - N$, so $\dot{k} = s_k k^\alpha h^\beta - Nk$; likewise $\dot{h} = s_h k^\alpha h^\beta - Nh$.

Step — steady state. $s_k y = Nk$ and $s_h y = Nh$ give $h/k = s_h/s_k$. Substituting into the first: $s_k k^\alpha (k s_h/s_k)^\beta = Nk \Rightarrow k^{1-\alpha-\beta} = s_k^{1-\beta} s_h^\beta / N$:

$$k^* = \left(\frac{s_k^{1-\beta} s_h^\beta}{N}\right)^{\frac{1}{1-\alpha-\beta}}, \qquad h^* = \left(\frac{s_k^\alpha s_h^{1-\alpha}}{N}\right)^{\frac{1}{1-\alpha-\beta}}.$$

Step — output. $\ln y^* = \alpha \ln k^* + \beta \ln h^* = \dfrac{\alpha \ln s_k + \beta \ln s_h - (\alpha+\beta)\ln N}{1-\alpha-\beta}$. Elasticity to s_k: $\alpha/(1-\alpha-\beta)$, which is 1 with $\alpha = \beta = 1/3$ (against 0.5 in the textbook Solow model).

Ref: Romer (2012), problems 4.5–4.9, pp. 185–186; Kurlat (2020), §5.2.

Grading rubric.

- Laws of motion with $N = n + g + \delta$: 1 pt.
- Steady state: 1 pt.
- $\ln y^*$ and elasticity: 1 pt.

b) (4 points) Prove local stability. Linearize the system in $x = \ln k$, $z = \ln h$, compute the Jacobian at the steady state, its eigenvalues, and the speed at which $\ln y$ converges. Compare with the textbook Solow model ($\beta = 0$) for $\alpha = \beta = 1/3$, $N = 0.06$.

Step — system in logs. $\dot{x} = s_k e^{(\alpha-1)x+\beta z} - N$, $\dot{z} = s_h e^{\alpha x+(\beta-1)z} - N$. At the steady state each exponential term equals N.

Step — Jacobian. Differentiating, $J = N \begin{pmatrix} \alpha - 1 & \beta \\ \alpha & \beta - 1 \end{pmatrix}$, with $\operatorname{tr} J = N(\alpha + \beta - 2) < 0$ and $\det J = N^2[(\alpha - 1)(\beta - 1) - \alpha\beta] = N^2(1 - \alpha - \beta) > 0$.

Step — eigenvalues. Discriminant $(\operatorname{tr} J)^2 - 4 \det J = N^2(\alpha + \beta)^2 \geq 0$, so the roots are real:

$$\mu_{1,2} = \tfrac{N}{2}\left[(\alpha + \beta - 2) \pm (\alpha + \beta)\right] \in \{-N(1 - \alpha - \beta),\ -N\}.$$

Both are negative: the steady state is a locally stable node.

Step — speed. $\ln y = \alpha x + \beta z$ converges at the slow root, $\lambda = (1 - \alpha - \beta)N = 2\%$ a year: half-life $\ln 2/0.02 = 34.7$ years. Textbook Solow: $(1 - \alpha)N = 4\%$, half-life 17.3 years.

Economic intuition. A broad capital share $\alpha + \beta = 2/3$ weakens diminishing returns, so convergence is slow, in line with the 2% estimates of cross-country regressions.

Ref: Romer (2012), §1.5, pp. 24–26, and problem 4.8.

Grading rubric.

- Log system and Jacobian: 1.5 pts.
- Trace/determinant argument for stability: 1 pt. Stability asserted without it: 0 for this line.
- Eigenvalues and speed: 1 pt.
- Comparison with Solow: 0.5 pt.

c) (4 points) Find the saving rates (s_k, s_h) that maximize steady-state consumption per effective worker $c^* = (1 - s_k - s_h)y^*$, and prove that they are a global maximum. What happens to the convergence speed and to y^* as $\alpha + \beta \to 1$, and what is the case $\beta \to 0$?

Step — objective. $\ln c^* = \ln(1 - s_k - s_h) + \dfrac{\alpha \ln s_k + \beta \ln s_h}{1 - \alpha - \beta} - \dfrac{\alpha + \beta}{1 - \alpha - \beta} \ln N$.

Step — FOCs. $\dfrac{-1}{1 - s_k - s_h} + \dfrac{\alpha}{(1 - \alpha - \beta)s_k} = 0$ and $\dfrac{-1}{1 - s_k - s_h} + \dfrac{\beta}{(1 - \alpha - \beta)s_h} = 0$. Dividing, $s_k/s_h = \alpha/\beta$. Then $(1 - \alpha - \beta)s_k = \alpha(1 - s_k - \frac{\beta}{\alpha}s_k) = \alpha - (\alpha + \beta)s_k$, so $s_k = \alpha$ and $s_h = \beta$.

Step — global maximum. $\ln(1 - s_k - s_h)$ is concave (log of an affine function) and $\alpha \ln s_k + \beta \ln s_h$ is concave, so $\ln c^*$ is concave on the open simplex and the FOC point is its global maximum. Equivalently, at $s_k = \alpha$ the net marginal product equals N: $f_k = \alpha y^*/k^* = \alpha N/s_k = N$.

Step — limits. As $\alpha + \beta \to 1$: $\lambda = (1 - \alpha - \beta)N \to 0$ and the exponent $1/(1 - \alpha - \beta) \to \infty$; the model becomes AK, where saving rates change growth rates, not levels. As $\beta \to 0$: the Solow model, with $s_{gr} = \alpha$.

Ref: Romer (2012), problem 1.5; Kurlat (2020), §4.3, eq. (4.3.4), pp. 61–62.

Grading rubric.

- Objective in logs and FOCs: 1.5 pts.
- Solution $s_k = \alpha$, $s_h = \beta$: 1 pt.
- Global-maximum argument: 1 pt. Only FOCs, no concavity: 0 for this line.
- Limits: 0.5 pt.

d) (4 points) Use $\alpha = \beta = 1/3$; Brazil: $s_k = 0.18$, $s_h = 0.10$, $N = 0.065$; US: $s_k = 0.22$, $s_h = 0.12$, $N = 0.06$. Compute the predicted ratio y^*_{BR}/y^*_{US}. Output per worker is $A \cdot y$; if the observed ratio of output per worker is 0.25, compute the ratio of A and of Hicks-neutral TFP $B \equiv A^{1-\alpha-\beta}$. Split the log gap into accumulation and technology, and say what the model predicts for the gap's dynamics.

Step — predicted ratio. With $\alpha = \beta = 1/3$ the coefficients in $\ln y^*$ are $1, 1, -2$: $\ln \frac{0.18}{0.22} + \ln \frac{0.10}{0.12} - 2 \ln \frac{0.065}{0.06} = -0.2007 - 0.1823 - 0.1601 = -0.5431$, ratio 0.581.

Step — technology. $\dfrac{A_{BR}}{A_{US}} = \dfrac{0.25}{0.581} = 0.430$; Hicks-neutral $B_{BR}/B_{US} = 0.430^{1/3} = 0.755$. The two numbers describe the same gap under different conventions.

Step — split. $\ln 0.25 = -1.386$: accumulation -0.543 (39%), technology -0.843 (61%).

Step — dynamics. Conditional convergence only: each country approaches its own y^* at 2% a year (half-life 35 years), but the 61% due to A never closes inside the model. Both countries save far below the Golden Rule ($s_k + s_h = 0.28 < 2/3$), so raising saving costs consumption during the transition.

Economic intuition. Human capital doubles the leverage of saving rates on income, yet even this generous model leaves most of Brazil's gap to technology and efficiency.

Ref: Romer (2012), §4.1–4.2; Kurlat (2020), §5.3–5.5, pp. 80–95.

Grading rubric.

- Predicted ratio: 1 pt.
- A and B ratios with the convention explained: 1 pt.
- Log split: 1 pt.
- Dynamics and Golden Rule reading: 1 pt. Claiming absolute convergence: 0 for this line.